Display Data: Box Plots

Lesson 8-5

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box from Q1 to Q3 with a line at the median, and whiskers from min to max

Box Plot

A graph that shows how data is spread using five key numbers.

Data: 10, 15, 20, 25, 30 → median is 20 (the 3rd value)

Median

The middle number that splits the data in half.

Data: 2, 4, 6, 8, 10, 12, 14 → Q1 = 4 (median of lower half), Q3 = 12 (median of upper half)

Quartile

Numbers that split the data into four equal parts.

If Q1 = 12 and Q3 = 20, then IQR = 20 − 12 = 8 — the middle 50% spans 8 units

Interquartile Range

The spread of the middle half of the data (Q3 – Q1).

A box plot where the median is centered in the box shows symmetric distribution

Data distribution

How the data looks: where it sits and how spread out it is.

Small IQR = low variability in the middle 50%. Large IQR = high variability

Variability

How spread out the numbers are.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can make and read a box plot to summarize a data set.

1️⃣ Notice

The basketball league wants to compare scoring across teams.

2️⃣ Key idea

I can make and read a box plot to summarize a data set.

3️⃣ Words I'll use

Box Plot **Median**

Quartile

Interquartile Range

Data distribution

Variability

🤔 Think About It

- What are the lowest and highest scores?
- What score is in the middle of the data?
- Are the scores bunched together or spread out?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The Wildcats' scores are 4, 6, 8, 10, 12, 14, 15, 18, 22, 24, 28. To build a box plot you need a five-number summary. What five numbers will you need?

Level 1 support

Start here: A box plot is built from the minimum, Q1, median, Q3, and maximum.

💡 **Need a hint?**

Sentence starters (optional)

A box plot needs the ____, Q1, median, Q3, and ____.

Un diagrama de caja necesita el ____, Q1, mediana, Q3 y el ____.

The median of these scores is ____.

La mediana de estos puntajes es ____.

Word bank: box plot | median | quartile | interquartile range | minimum

Listen for: *Students name the five-number summary (min, Q1, median, Q3, max) and find the median (14) for the 11 ordered scores.*

Level 2

Justify: explain how to find Q1 and Q3 for this data set and what they represent.

- Q1 is the median of the ____ half, which is ____.
- Q3 is the median of the ____ half, which is ____.

EXPLORE

As you draw the box from Q1 to Q3, what does the length of the box tell you about the players' scores?

Level 1 support

Start here: The box spans the middle 50% of the data; its length is the interquartile range.

💡 **Need a hint?**

Sentence starters (optional)

The box covers the middle ____ of the scores.

La caja cubre el ____ medio de los puntajes.

A longer box means the middle scores are ____.

Una caja más larga significa que los puntajes del medio están ____.

Word bank: **box plot** **median** **quartile** **interquartile range** **spread**

Listen for: *A strong answer explains the box covers the middle 50% (Q1 to Q3), its length is the IQR, and a longer box means greater spread.*

Level 2

Compare: how would the box plot change if the highest score were 50 instead of 28? Which part changes, the box or the whisker?

- The ____ would change because ____.
- The box would stay about the same since ____.

CONNECT

Team A: Min=40, Q1=52, Med=60, Q3=68, Max=75. Team B: Min=35, Q1=55, Med=61, Q3=65, Max=80. A fan says Team B is better because they scored 80 once. Which team scores high more consistently?

Level 1 support

Start here: A single high value matters less than the median and the spread of the middle 50%.

💡 **Need a hint?**

Sentence starters (optional)

Team ___ scores high more consistently because their ___ is higher.

El Equipo ___ anota alto más consistentemente porque su ___ es mayor.

The 80-point game is just the ___, not the typical score.

El juego de 80 puntos es solo el ___, no el puntaje típico.

Word bank: box plot | median | quartile | interquartile range | maximum

Listen for: *Students compare medians and Q1/Q3 (Team B has a higher Q1 and median) and explain the single max of 80 doesn't make a team consistently better.*

Level 2

Critique: the fan focuses only on the maximum. What two box-plot features give a fairer comparison? Defend your choice.

- A fairer comparison looks at the ___ and the ___.
- The maximum alone is misleading because ___.

REFLECT

A box plot has Q1 = 20 and Q3 = 36. What is the IQR, and what does it represent?

Level 1 support

Start here: The IQR is Q3 minus Q1 and measures the spread of the middle 50% of the data.

💡 **Need a hint?**

Sentence starters (optional)

The IQR is ___ because Q3 minus Q1 equals ___.

El RIC es ___ porque Q3 menos Q1 es ___.

It represents the spread of the middle ___ of the data.

Representa la dispersión del ___ medio de los datos.

Word bank: interquartile range | quartile | median | box plot | spread

Listen for: *Students compute IQR = 36 - 20 = 16 and explain it is the range of the middle 50% of the data.*

Level 2

Generalize: why is the IQR a more reliable measure of spread than the full range for a box plot with an extreme value?

- The IQR is more reliable because it ignores ___.
- The full range can be misleading when ___.

Guided Examples

Example 1

A box plot shows: Min = 10, Q1 = 15, Median = 20, Q3 = 28, Max = 35. What is the interquartile range (IQR)?

Solution: IQR = Q3 − Q1 = 28 − 15 = 13.

Answer: A. 13

Example 2

On a box plot, what does the line inside the box represent?

Solution: The line inside the box of a box plot always represents the median.

Answer: A. The median

Example 3

What percentage of data falls between Q1 and Q3 on a box plot?

Solution: The box in a box plot contains the middle 50% of the data (from Q1 to Q3).

Answer: A. 50%

Write About the Math

The Writing Revolution

I can explain my box plot using the words box plot, median, quartile, and interquartile range.

1. Kernel Sentence

subject + verb

Model: Box Plot is a graph that shows how data is spread using five key numbers.
Diagrama de caja es una gráfica que muestra cómo se reparten los datos con cinco números clave.

Write a kernel sentence about box plot. Use a subject and a verb.
Escribe una oración base sobre diagrama de caja. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Box Plot matters in math
Diagrama de caja importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Box Plot matters in math because ____.
Diagrama de caja importa en matemáticas porque ____.

____

but
pero

Box Plot matters in math, but ____.
Diagrama de caja importa en matemáticas, pero ____.

____

so
entonces

Box Plot matters in math, so ____.
Diagrama de caja importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about box plot.
Di un hecho verdadero sobre box plot.

Box plot ____.

Question
Pregunta

Ask a question about box plot.
Haz una pregunta sobre box plot.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about box plot.
Muestra entusiasmo sobre box plot.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with box plot.
Dile a un compañero qué hacer con box plot.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

A box plot needs the ____, Q1, median, Q3, and ____.
Un diagrama de caja necesita el ____, Q1, mediana, Q3 y el ____.

The median of these scores is ____.
La mediana de estos puntajes es ____.

The box covers the middle ____ of the scores.
La caja cubre el ____ medio de los puntajes.

Try It

Solve on your own. Check the answer key when you are done.

1. Find the range of: 14, 22, 8, 31, 17

A. 23

B. 31

C. 8

D. 18

Show your work:

2. Two box plots show: Team A has IQR = 8, Team B has IQR = 22. What does this tell you?

A. Team A's middle 50% is more consistent; Team B's is more spread out

B. Team B is a better team

C. Team A has a higher median

D. The IQRs don't tell you about consistency

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two basketball teams played 9 games each. Team A: 40, 42, 44, 46, 48, 50, 52, 54, 56. Team B: 30, 35, 40, 45, 48, 50, 55, 60, 65. Find the five-number summary and IQR for each team. Which team is more consistent? Which has a wider spread in the middle 50%?

Sentence starter: Team A: Min=____, Q1=____, Median=____, Q3=____, Max=____, IQR=____. Team B: Min=____, Q1=____, Median=____, Q3=____, Max=____, IQR=____. Team ____ is more consistent because its IQR is ____.

Show your work:

Reflect — Exit Ticket

A box plot has Q1 = 20 and Q3 = 36. What is the IQR and what does it represent?

A. IQR = 16; it is the spread of the middle 50% of the data

B. IQR = 56; it is the total of Q1 and Q3

C. IQR = 28; it is the median between Q1 and Q3

D. IQR = 36; it is the value of Q3

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 23 *— Range = max − min = 31 − 8 = 23.*
2. **Try It 2:** A. Team A's middle 50% is more consistent; Team B's is more spread out *— A smaller IQR means the middle 50% of data is more tightly clustered. Team A's scoring is more consistent in the middle range.*
3. **Exit Ticket:** A. IQR = 16; it is the spread of the middle 50% of the data *— IQR = Q3 − Q1 = 36 − 20 = 16. The IQR tells you the range of the middle 50% of the data.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Box Plot is a graph that shows how data is spread using five key numbers.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).